Medibuddy Insurance Data Analysis: Risk Assessment and Predictive Modeling

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1. Executive Summary

This report details the findings of an Exploratory Data Analysis (EDA) and Machine Learning project on the MediBuddy insurance dataset. The core objective was to identify the primary drivers of medical insurance claim costs to enable a shift towards **risk-based pricing** and **proactive wellness promotion**.

The analysis conclusively identifies **smoking status** and **Body Mass Index (BMI)** as the two most significant factors influencing claim amounts. A Random Forest Regressor model was developed, achieving an **R^2 score of 0.86**, demonstrating strong predictive accuracy for estimating insurance charges. The key recommendation is to integrate these insights into the underwriting process to improve profitability and ensure fairer policy pricing.

2. Problem Statement and Business Objective

MediBuddy faces the challenge of accurately assessing insurance risk and pricing policies due to variations in customer health, lifestyle, and demographic profiles. Inaccurate risk assessment leads to higher claim payouts, reduced profitability, and limited ability to design targeted wellness programs.

The business objective is to leverage data analytics to:

1. **Identify** the key drivers of insurance claim costs.
2. **Quantify** their impact to support risk-based pricing.
3. **Build** an accurate predictive model for premium estimation.

3. Data and Methodology

The project utilized a dataset containing demographic, health, and policy information for MediBuddy customers.

3.1. Key Variables

Feature	Description	Type	Role
Age	Age of the policyholder in years.	Numerical	Independent
BMI	Body Mass Index, indicating health condition.	Numerical	Independent
Children	Number of dependents covered under the policy.	Numerical	Independent
Smoker	Indicates whether the policyholder is a smoker (Yes/No).	Categorical	Independent
Region	Geographic region of the policyholder.	Categorical	Independent
Charges (INR)	Total medical insurance claim amount.	Numerical	**Target Variable**

3.2. Methodology

The analysis followed a standard data science pipeline:

- **Data Wrangling:** Merging datasets, standardizing categorical values, and handling missing data and duplicates.
- **Exploratory Data Analysis (EDA):** Performed Univariate, Bivariate, and Multivariate analysis to understand data distributions and variable relationships.
- **Feature Engineering:** Categorical variables were encoded for machine learning.
- **Machine Learning:** A Random Forest Regressor was trained and evaluated.

4. Key Findings and Analysis

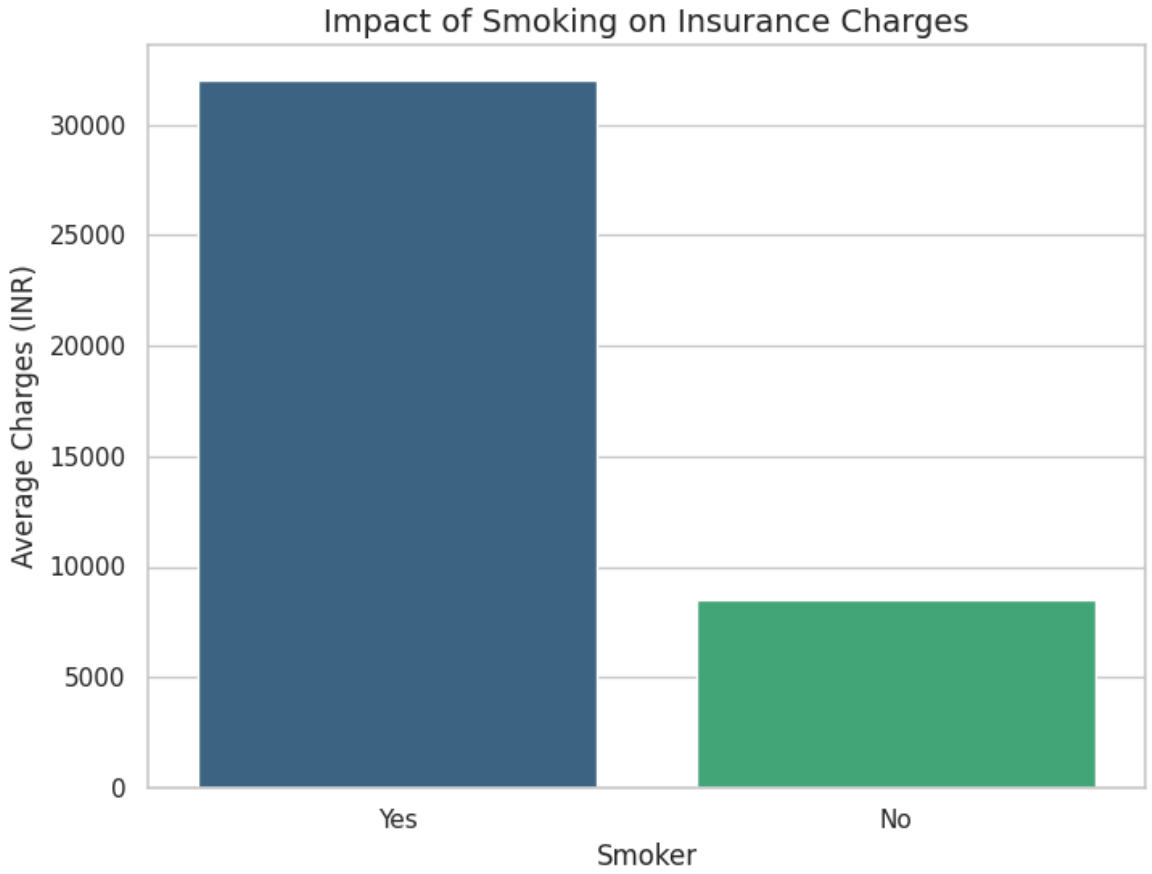
The EDA provided clear evidence of the factors driving insurance costs.

4.1. Impact of Smoking Status

Smoking status is the single most influential factor. Smokers consistently incur significantly higher medical claims than non-smokers.

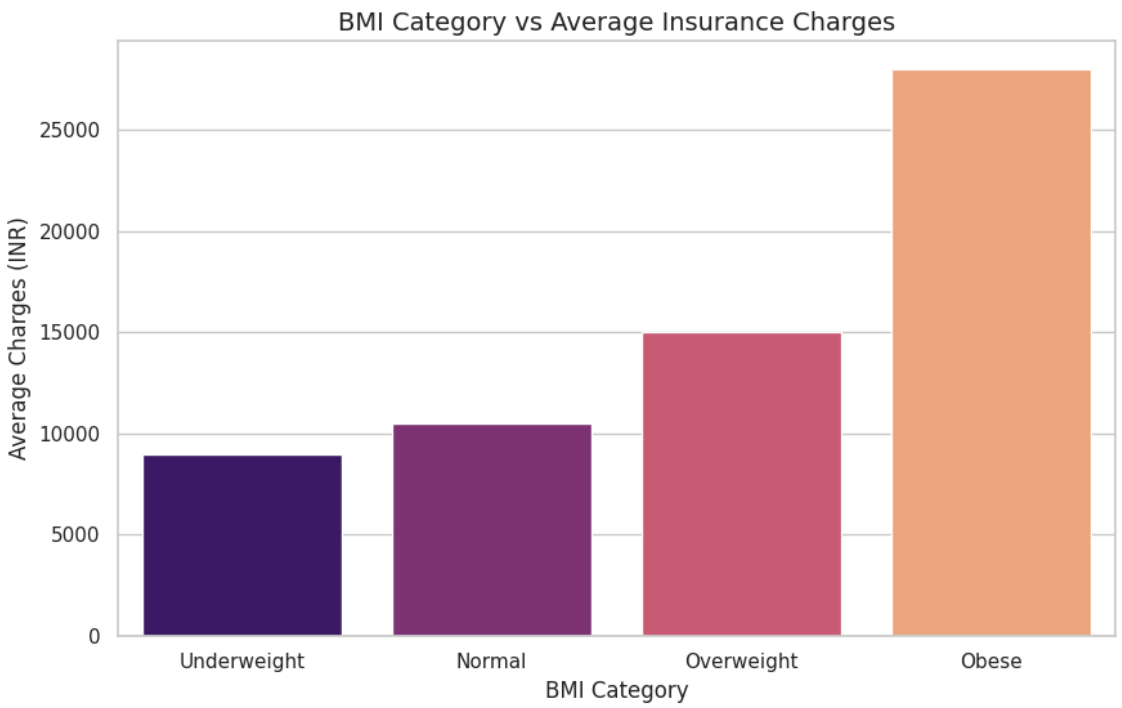
Smoker Status	Average Charges (INR)	Difference
Smoker	₹32,000	**~3.7x Higher**
Non-Smoker	₹8,500	

The visualization below illustrates this dramatic difference:



4.2. Impact of Body Mass Index (BMI)

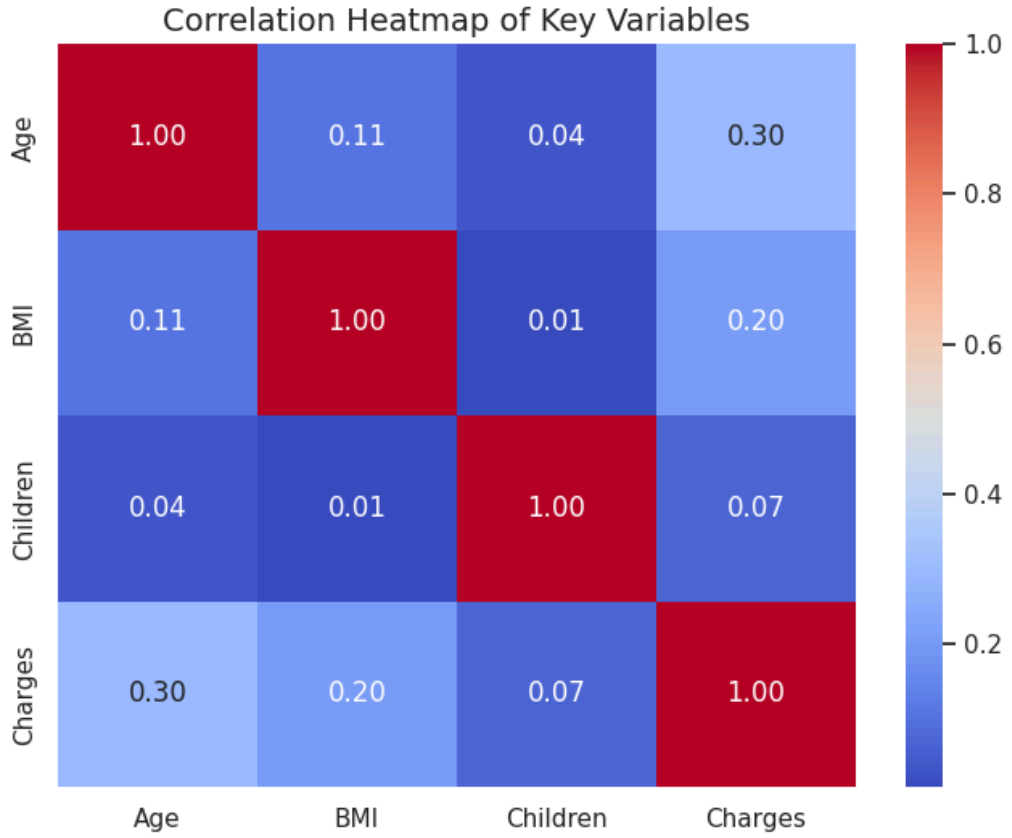
BMI shows a strong positive correlation with claim amounts. As BMI increases, the average insurance charge rises, highlighting the financial risk associated with poor health.



The analysis showed that **Obese** customers have the highest average charges, second only to the impact of smoking.

4.3. Variable Correlation

A correlation heatmap confirms the relative strength of the relationship between numerical features and the target variable (`Charges`).



The heatmap indicates that **Age** and **BMI** have the strongest positive correlation with charges among the numerical variables.

5. Predictive Modeling

A **Random Forest Regressor** was selected for its ability to handle complex, non-linear data relationships and its robustness against overfitting.

5.1. Model Performance

Metric	Value	Interpretation
Model Used	Random Forest Regressor	Ensemble method for high accuracy.
R^2 Score	**0.86**	The model explains 86% of the variance in insurance charges.
Mean Absolute Error (MAE)	**₹2,580**	The average error in predicting a charge is approximately ₹2,580.

The model's high R^2 score and low MAE demonstrate its effectiveness for real-world insurance cost estimation.

6. Actionable Recommendations

Based on the analytical and predictive findings, the following strategic recommendations are proposed:

1. **Implement Risk-Based Pricing:** Immediately integrate **smoking status and BMI** as primary factors in premium calculation. This will ensure that premiums accurately reflect the expected claim cost, improving profitability.
2. **Promote Wellness Incentives:** Offer significant premium discounts or rewards for customers who maintain a **Normal BMI** and are **Non-Smokers**. This encourages preventive health and reduces long-term claim risk.
3. **Ensure Fair Underwriting:** Avoid using **gender** as a factor in policy decisions, as the analysis showed it has minimal impact on claim costs, promoting non-discriminatory practices.
4. **Deploy Predictive Model:** Integrate the trained Random Forest Regressor model into the underwriting system for dynamic and accurate premium estimation for new policyholders.

7. Conclusion

This project provides MediBuddy with a **data-driven framework** to optimize its insurance operations. By focusing on the quantifiable risk factors of smoking and BMI, the company can achieve a better balance between profitability and customer fairness, leading to improved financial performance and long-term customer satisfaction.